

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

Sets

Section 2.1

2.1.4 Set Cardinality

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* (基数) of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

2.1.5 Power Sets

Definition: The set of all subsets of a set A , denoted $P(A)$, is called the *power set* (幂集) of A .

Example: If $A = \{a, b\}$ then

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

If a set has n elements, then the cardinality of the power set is 2^n . (In Chapters 5 and 6, we will discuss different ways to show this.)

2.1.6 Tuples

The *ordered n -tuple* (有序 n 元组) $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.

Two n -tuples are equal if and only if their corresponding elements are equal.

2-tuples are called *ordered pairs* (有序对).

The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Cartesian Product₁

René Descartes
(1596-1650)



Definition: The *Cartesian Product* (笛卡尔积, 集合的直积) of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$$

Example:

$$A = \{a,b\} \quad B = \{1,2,3\}$$

$$A \times B = \{(a,1),(a,2),(a,3), (b,1),(b,2),(b,3)\}$$

Definition: A subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B . (Relations will be covered in depth in Chapter 9.)

Cartesian Product₂

Definition: The **Cartesian products** of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n =$$

$$\{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0, 1\}$, $B = \{1, 2\}$ and $C = \{0, 1, 2\}$

Solution: $A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}$

We use the notation A^2 to denote $A \times A$, the Cartesian product of the set A with itself. Similarly, $A^3 = A \times A \times A$, $A^4 = A \times A \times A \times A$, and so on. More generally,

$$A^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A \text{ for } i = 1, 2, \dots, n\}.$$

Suppose that $A = \{1, 2\}$.

$$A^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

$$A^3 = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}.$$

Section Summary₁

Definition of sets

Describing Sets

- Roster Method $S = \{a, b, c, d\}$
- Set-Builder Notation $S = \{x \mid x \text{ is a positive integer less than } 100\}$

Some Important Sets in Mathematics

Empty Set and Universal Set

Subsets and Set Equality

Cardinality of Sets: is the number of (distinct) elements of A , $|A|$.

Power Set: If $A = \{a, b\}$ then $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Tuples: the *ordered n -tuple* (有序 n 元组) $(a_1, a_2, \dots, a_n)$

Cartesian Product $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$

Exercise

21. What is the cardinality of each of these sets?

a) $\{a\}$

b) $\{\{a\}\}$

c) $\{a, \{a\}\}$

d) $\{a, \{a\}, \{a, \{a\}\}\}$

23. Find the power set of each of these sets, where a and b are distinct elements.

a) $\{a\}$

b) $\{a, b\}$

c) $\{\emptyset, \{\emptyset\}\}$

29. Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find

a) $A \times B$.

b) $B \times A$.

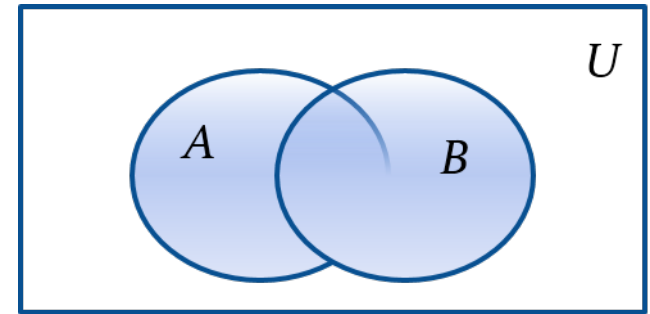
Set Operations

Section 2.2

Union

Definition: Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set:

$$\{x \mid x \in A \vee x \in B\}$$

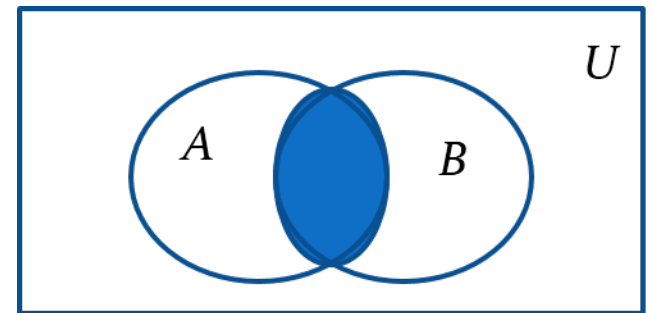


Venn Diagram for $A \cup B$

Intersection

Definition: The *intersection* of sets A and B , denoted by $A \cap B$, is $\{x \mid x \in A \wedge x \in B\}$

Note if the intersection is empty, then A and B are said to be *disjoint*.



Venn Diagram for $A \cap B$

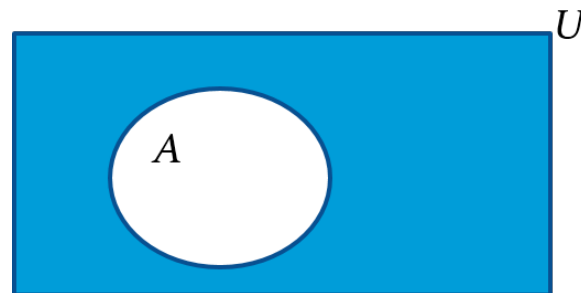
Complement

Definition: If A is a set, then the *complement* of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c .)

Venn Diagram for Complement

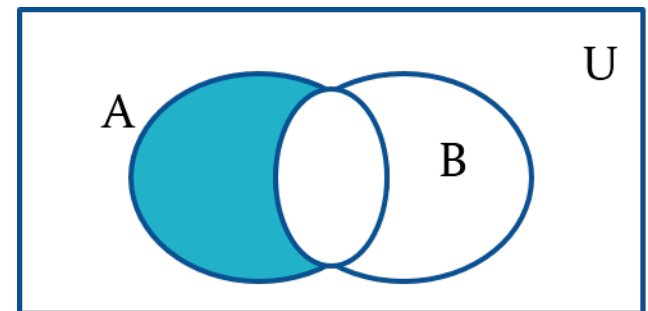


Difference

Definition: Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing the elements of A that are not in B . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \bar{B}$$

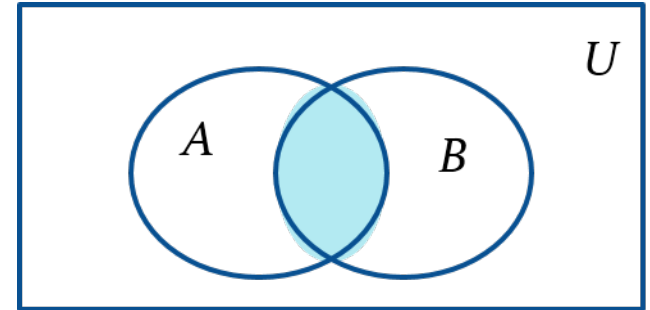
Venn Diagram for $A - B$



The Cardinality of the Union of Two Sets

Inclusion-Exclusion(容斥原理)

$$|A \cup B| = |A| + |B| - |A \cap B|$$



Venn Diagram for A , B , $A \cap B$, $A \cup B$

2.2.2 Set Identities₁

Identity laws (恒等律)

$$A \cup \emptyset = A \quad A \cap U = A$$

Domination laws (支配律)

$$A \cup U = U \quad A \cap \emptyset = \emptyset$$

Idempotent laws (幂等律)

$$A \cup A = A \quad A \cap A = A$$

Complementation law

$$\overline{\overline{A}} = A$$

Set Identities₂

Commutative laws (交换律)

$$A \cup B = B \cup A \qquad A \cap B = B \cap A$$

Associative laws (结合律)

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive laws (分配律)

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Set Identities₃

De Morgan's laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Absorption laws (吸收律)

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Complement laws

$$A \cup \bar{A} = U$$

$$A \cap \bar{A} = \emptyset$$

Proving Set Identities

Different ways to prove set identities:

1. Prove that each set (side of the identity) is a subset of the other.
2. Use **set builder notation** and propositional logic.
3. **Membership Tables** : Verify that elements in the same combination of sets always either belong or do not belong to the both side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not

Proof of Second De Morgan Law

Example: Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Method 1: We prove this identity by showing that:

$$1) \overline{A \cap B} \subseteq \overline{A} \cup \overline{B} \quad \text{and}$$

$$2) \overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$$

Proof of Second De Morgan Law₂

These steps show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$	by assumption
$x \notin A \cap B$	defn. of complement
$\neg((x \in A) \wedge (x \in B))$	by defn. of intersection
$\neg(x \in A) \vee \neg(x \in B)$	1st De Morgan law for Prop Logic
$x \notin A \vee x \notin B$	defn. of negation
$x \in \overline{A} \vee x \in \overline{B}$	defn. of complement
$x \in \overline{A} \cup \overline{B}$	by defn. of union

Proof of Second De Morgan Law₃

These steps show that: $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$$x \in \overline{A} \cup \overline{B}$$

by assumption

$$(x \in \overline{A}) \vee (x \in \overline{B})$$

by defn. of union

$$(x \notin A) \vee (x \in \overline{B})$$

defn. of complement

$$\neg(x \in A) \vee \neg(x \in B)$$

defn. of negation

$$\neg((x \in A) \wedge \neg(x \in B))$$

1st De Morgan law for Prop Logic

$$\neg(x \in A \cap B)$$

defn. of intersection

$$x \in \overline{A \cap B}$$

defn. of complement

Method 2:Set-Builder Notation: Second De Morgan Law

$$\begin{aligned}\overline{A \cap B} &= x \in \overline{A \cap B} && \text{by defn. of complement} \\ &= \{x \mid \neg(x \in (A \cap B))\} && \text{by defn. of does not belong symbol} \\ &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by defn. of intersection} \\ &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by 1st De Morgan law for} \\ &&& \text{Prop Logic} \\ &= \{x \mid x \notin A \vee x \notin B\} && \text{by defn. of not belong symbol} \\ &= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{by defn. of complement} \\ &= \{x \mid x \in \overline{A} \cup \overline{B}\} && \text{by defn. of union} \\ &= \overline{A} \cup \overline{B} && \text{by meaning of notation}\end{aligned}$$

Method 3: Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

2.2.3 Generalized Unions and Intersections

Let $A_1, A_2, \dots, A_n$ be an indexed collection of sets.

We define:
$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

These are well defined, since union and intersection are associative.

For $i = 1, 2, \dots$, let $A_i = \{i, i + 1, i + 2, \dots\}$. Then,

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{1, 2, 3, \dots\}$$

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{n, n + 1, n + 2, \dots\} = A_n$$

Section Summary₂

Set Operations

- Union
- Intersection
- Complementation
- Difference

More on Set Cardinality $|A \cup B| = |A| + |B| - |A \cap B|$

Set Identities

Proving Identities

Membership Tables

Proving Set Identities

Different ways to prove set identities:

1. Prove that each set (side of the identity) is a subset of the other.
2. Use **set builder notation** and propositional logic.
3. **Membership Tables** : Verify that elements in the same combination of sets always either belong or do not belong to the both side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not

Exercise

15. Prove the second De Morgan law in Table 1 by showing that if A and B are sets, then $\overline{A \cup B} = \overline{A} \cap \overline{B}$

- a) by showing each side is a subset of the other side.
- b) using a membership table.

53. Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find

a) $\bigcup_{i=1}^n A_i.$

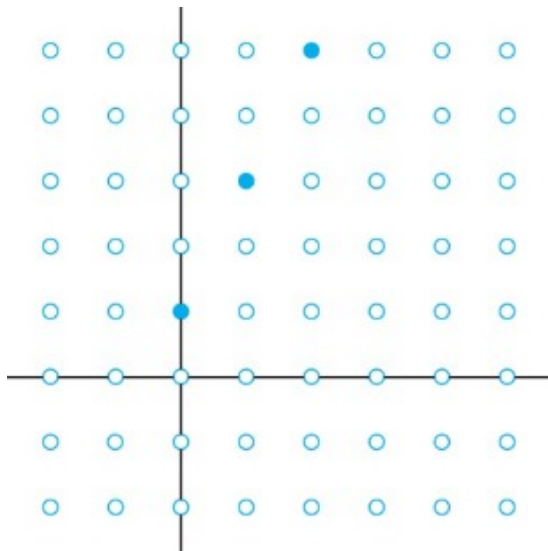
b) $\bigcap_{i=1}^n A_i.$

Functions

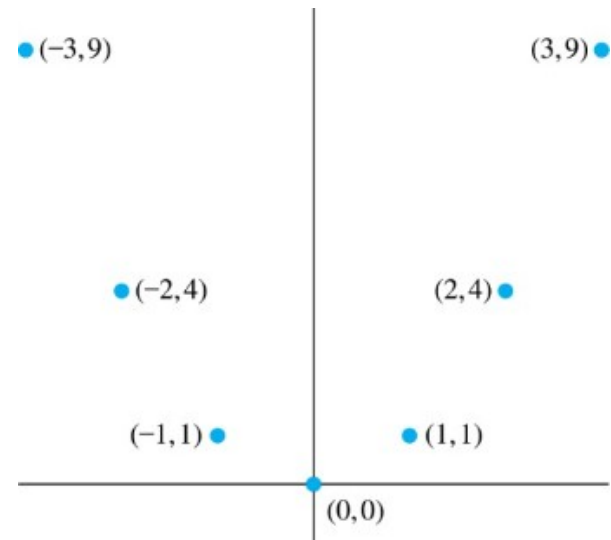
Section 2.3

2.3.4 Graphs of Functions

Let f be a function from the set A to the set B . The *graph* of the function f is the set of ordered pairs $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.



Graph of $f(n) = 2n + 1$
from $\mathbb{Z}$ to $\mathbb{Z}$



Graph of $f(x) = x^2$
from $\mathbb{Z}$ to $\mathbb{Z}$

[Jump to long description](#)

2.3.5 Some Important Functions

The *floor* function(向下取整函数), denoted

$$f(x) = \lfloor x \rfloor$$

is the largest integer less than or equal to x .

The *ceiling* function(向上取整函数), denoted

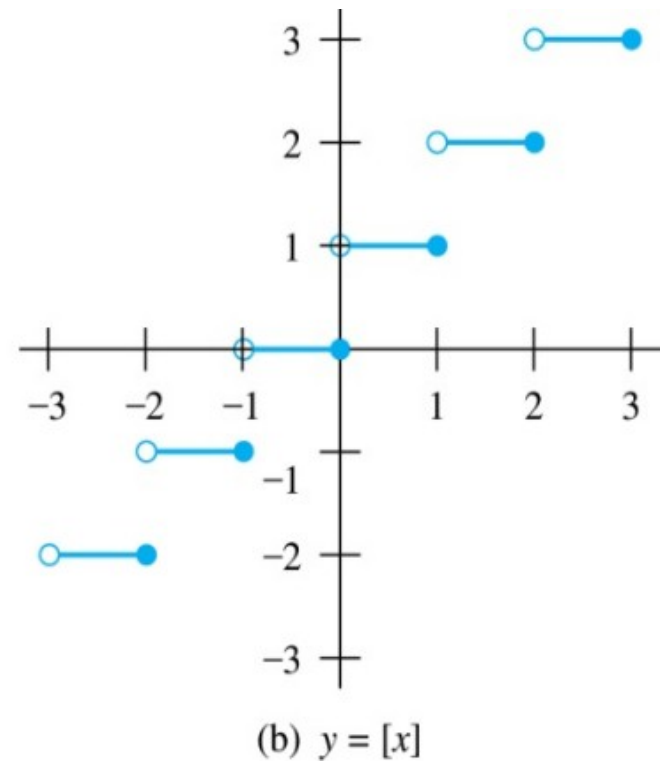
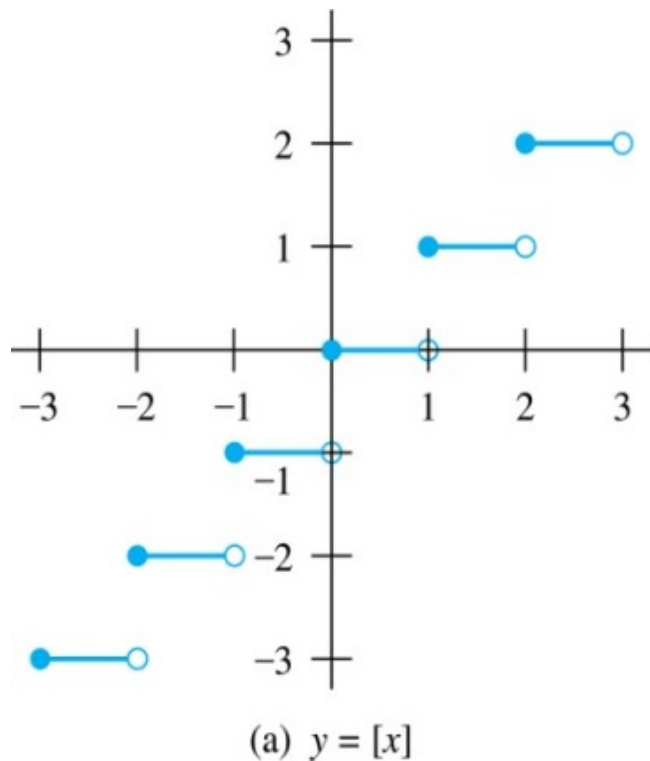
$$f(x) = \lceil x \rceil$$

is the smallest integer greater than or equal to x

Example:

$$\begin{array}{ll} \lceil 3.5 \rceil = 4 & \lfloor 3.5 \rfloor = 3 \\ \lceil -1.5 \rceil = -1 & \lfloor -1.5 \rfloor = -2 \end{array}$$

Floor and Ceiling Functions₁



Graph of (a) Floor and (b) Ceiling Functions

[Jump to long description](#)

Factorial Function(阶乘函数)

Definition: $f: \mathbf{N} \rightarrow \mathbf{Z}^+$, denoted by $f(n) = n!$ is the product of the first n positive integers when n is a nonnegative integer.

$$f(n) = 1 \cdot 2 \cdots (n-1) \cdot n, \quad f(0) = 0! = 1$$

Examples:

$$f(1) = 1! = 1$$

$$f(2) = 2! = 1 \cdot 2 = 2$$

$$f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$$

$$f(20) = 2,432,902,008,176,640,000.$$

Section Summary₃

Definition of a Function.

- Domain, Codomain, Range
- Image, Preimage

Injection, Surjection, Bijection

Inverse Function

Function Composition

Graphing Functions

Floor, Ceiling, Factorial

Exercise

Find the values of the following expressions.

8. Find these values.

a) $\lfloor 1.1 \rfloor$

b) $\lceil 1.1 \rceil$

c) $\lfloor -0.1 \rfloor$

d) $\lceil -0.1 \rceil$

Sequences and Summations

Section 2.4

2.4.2 Sequences₁

Definition: A *sequence*(序列) is a function from a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) to a set S .

We call a_n a *term*(项) of the sequence.

Sequences₂

Example: Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n} \qquad \{a_n\} = \{a_1, a_2, a_3 \dots\}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$$

Geometric Progression

Definition: A *geometric progression*(几何级数 等比级数) is a sequence of the form: $a, ar^2, \dots, ar^n, \dots$ where the *initial term*(初始项) a and the *common ratio*(公比) r are real numbers.

Arithmetic Progression

Definition: A *arithmetic progression*(算术级数 等差级数) is a sequence of the form: $a, a + d, a + 2d, \dots, a + nd, \dots$ where the *initial term* a and the *common difference*(公差) d are real numbers.

2.4.3 Recurrence Relations

Definition: A *recurrence relation*(递推关系) for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.

The *initial conditions*(初始条件) for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Fibonacci Sequence

Definition: Define the *Fibonacci sequence*(斐波那契数列), $f_0, f_1, f_2, \dots$, by:

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example: Find f_2, f_3, f_4, f_5 and f_6 .

Answer :

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$



Solving Recurrence Relations

Finding a formula for the n th term of the sequence generated by a recurrence relation is called *solving the recurrence relation*.

Such a formula is called a *closed formula*(封闭公式, 通项).

Iterative Solution Example₁

Method 1: Working upward, forward substitution Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$a_2 = 2 + 3$$

$$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

.

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$$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$$

2.4.5 Summations₁

Sum of the terms

$$a_m, a_{m+1}, \dots, a_n$$

from the sequence $\{a_n\}$. We use the notation

$$\sum_{j=m}^n a_j, \quad \sum_{j=m}^n a_j, \quad \text{or} \quad \sum_{m \leq j \leq n} a_j$$

(read as the sum from $j = m$ to $j = n$ of a_j) to represent

$$a_m + a_{m+1} + \dots + a_n.$$

The variable j is called the *index of summation*. It runs through all the integers starting with its *lower limit* m and ending with its *upper limit* n .

Geometric Series₁

Sums of terms of geometric progressions

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & r \neq 1 \\ (n+1)a & r = 1 \end{cases}$$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, \ r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=0}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Geometric Series:
We just proved this.

Later we will
prove some
of these by
induction.

Proof in text
(requires calculus)

Section Summary⁴

Sequences.

- Examples: Geometric Progression, Arithmetic Progression

Recurrence Relations

- Example: Fibonacci Sequence $f_0 = 0, f_1 = 1 : f_n = f_{n-1} + f_{n-2}$

Summations

Exercise

32. Find the value of each of these sums.

b) $\sum_{j=0}^8 (3^j - 2^j)$

Matrices

Section 2.6

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding *entries* in every position are equal.

$$3 \times 2 \text{ matrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$

Notation

Let m and n be positive integers and let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The i th row of $\mathbf{A}$ is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$. The j th column of $\mathbf{A}$ is the $m \times 1$ matrix:

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \cdot \\ \cdot \\ a_{mj} \end{bmatrix}$$

The (i,j) th *element* or *entry* of $\mathbf{A}$ is the element a_{ij} . We can use $\mathbf{A} = [a_{ij}]$ to denote the matrix with its (i,j) th element equal to a_{ij} .

2.6.2 Matrix Arithmetic: Addition

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note that matrices of different sizes can not be added.

Matrix Multiplication

Definition: Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$. In other words, if $\mathbf{AB} = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$.

Illustration of Matrix Multiplication

The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \mathbf{a_{i1}} & \mathbf{a_{i2}} & \dots & \mathbf{a_{ik}} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & a_{12} & \dots & \mathbf{b_{1j}} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & \mathbf{b_{2j}} & \dots & b_{2n} \\ \cdot & \cdot & & \cdot & & \cdot \\ \cdot & \cdot & & \cdot & & \cdot \\ b_{k1} & b_{k2} & \dots & \mathbf{b_{kj}} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & b_{22} & \dots & c_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \mathbf{c_{ij}} & \cdot \\ \cdot & \cdot & & \cdot \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$\mathbf{c_{ij}} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Matrix Multiplication is not Commutative

Example: Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Does **$AB = BA$** ?

Solution:

$$AB = \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

$AB \neq BA$

2.6.3 Transposes and Powers of Matrices

Identity Matrix and Powers of Matrices

Definition: The *identity matrix* of order n is the $m \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & & \cdot \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_m\mathbf{A} = \mathbf{A}$$

when $\mathbf{A}$ is an $m \times n$ matrix

Powers of square matrices can be defined. When $\mathbf{A}$ is an $n \times n$ matrix, we have:

$$\mathbf{A}^0 = \mathbf{I}_n$$

$$\mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A}\cdots\mathbf{A}}_{r \text{ times}}$$

Transposes of Matrices₁

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The *transpose* of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Transposes of Matrices₂

Definition: A square matrix $\mathbf{A}$ is called **symmetric** if $\mathbf{A} = \mathbf{A}^t$. Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.

The matrix $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is symmetric.

2.6.4 Zero-One Matrices₁

Definition: A matrix all of whose entries are either 0 or 1 is called a *zero-one matrix*(0-1矩阵). (These will be used in Chapters 9 and 10.)

Algorithms operating on discrete structures represented by zero-one matrices are based on **Boolean arithmetic** defined by the following **Boolean operations**(布尔运算):

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Zero-One Matrices₂

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ zero-one matrices.

- The *join*(并) of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. The *join* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The *meet*(交) of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \wedge \mathbf{B}$.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The join of **A** and **B** is

$$A \vee B = \begin{bmatrix} 1 & \vee & 0 & 0 & \vee & 1 & 1 & \vee & 0 \\ 0 & \vee & 1 & 1 & \vee & 1 & 0 & \vee & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

The meet of **A** and **B** is

$$A \wedge B = \begin{bmatrix} 1 & \wedge & 0 & 0 & \wedge & 1 & 1 & \wedge & 0 \\ 0 & \wedge & 1 & 1 & \wedge & 1 & 0 & \wedge & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₁

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The **Boolean product**(布尔积) of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Example: Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₂

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Solution: The Boolean product $A \odot B$ is given by

$$\begin{aligned} A \odot B &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Boolean Powers of Zero-One Matrices₁

Definition: Let $\mathbf{A}$ be a square zero-one matrix and let r be a positive integer. The r th Boolean power of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$, denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \dots \odot \mathbf{A}}_{r \text{ times}}.$$

We define $\mathbf{A}^{[0]}$ to be $\mathbf{I}_n$.

(The Boolean product is well defined because the Boolean product of matrices is associative.)

Boolean Powers of Zero-One Matrices₂

Example: Let

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find $A^{[n]}$ for all positive integers n .

Solution:

$$A^{[2]} = A \odot A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^{[3]} = A^{[2]} \odot A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[4]} = A^{[3]} \odot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[n]} = A^{[5]} \text{ for all positive integers } n \text{ with } n \geq 5$$

Section Summary₇

Definition of a Matrix

Matrix Arithmetic

Transposes and Powers of Arithmetic

Zero-One matrices

Boolean operations

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Boolean product $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry $c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj})$.

Exercise

20. Let

$$\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}.$$

b) Find $\mathbf{A}^3$.

26. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Find

a) $\mathbf{A} \vee \mathbf{B}$.

b) $\mathbf{A} \wedge \mathbf{B}$.

c) $\mathbf{A} \odot \mathbf{B}$.

28. Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Chapter Summary

Sets

- The Language of Sets
- Set Operations
- Set Identities

Functions

- Types of Functions
- Operations on Functions
- Computability

Sequences and Summations

- Types of Sequences
- Summation Formulae

Matrices

- Matrix Arithmetic

Proving Set Identities

Different ways to prove set identities:

1. Prove that each set (side of the identity) is a subset of the other.
2. Use **set builder notation** and propositional logic.
3. **Membership Tables** : Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity.
Use 1 to indicate it is in the set and a 0 to indicate that it is not

1, The *Cartesian Product* (笛卡尔积) of two sets A and B , $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$$

$$2 \quad A^0 = I_n \quad A^r = \underbrace{A A A \cdots A}_{r \text{ times}}$$

3 The *join*(并) of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. $\mathbf{A} \vee \mathbf{B}$. The *meet*(交) of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. $\mathbf{A} \wedge \mathbf{B}$.

4 Boolean product $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry $c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \cdots \vee (a_{ik} \wedge b_{kj})$.

$$5 \quad A^{[r]} = \underbrace{A \odot A \odot \cdots \odot A}_{r \text{ times}}$$

Homework

Page 132 34 a

Page 144 23 (1, Use set builder notation and propositional logic. 2, Membership Tables).

Page 164 65

Page 179 31 d.

Page 195 29

34. Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $C = \{0, 1\}$. Find

b) $C \times B \times A$.

21. What is the cardinality of each of these sets?

b) $\{\{a\}\}$

d) $\{a, \{a\}, \{a, \{a\}\}\}$

22. Prove that

12. Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.

9. Find these values.

a) $\lceil \frac{3}{4} \rceil$

b) $\lfloor \frac{7}{8} \rfloor$

c) $\lceil -\frac{3}{4} \rceil$

d) $\lfloor -\frac{7}{8} \rfloor$

31. What is the value of each of these sums of terms of a geometric progression?

a) $\sum_{j=0}^8 3 \cdot 2^j$

27. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Find

a) $\mathbf{A} \vee \mathbf{B}.$

b) $\mathbf{A} \wedge \mathbf{B}.$

c) $\mathbf{A} \odot \mathbf{B}.$